



Balancing Fairness and Performance Under Multiple Sensitive Attributes

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Abstract. Advances in machine learning enable solutions to increasingly complex problems. However, the predominant focus on predictive accuracy in many models often results in insufficient attention to potential biases against certain groups, thereby highlighting the critical need for fairness-aware machine learning. While most existing studies focus solely on debiasing with respect to a single sensitive attribute (e.g., race or gender), they fail to simultaneously consider fairness under multiple sensitive attributes. Furthermore, current fairness-enhancing approaches frequently degrade model performance. To address these limitations, we propose a novel framework named BFPM that achieves a better balance between fairness and performance across multiple sensitive attributes. BFPM consists of two parts. First, in the data pre-processing stage, we generate synthetic samples to balance the proportion of multiple sensitive attributes in the dataset, thereby enhancing fairness. Second, in the in-processing stage, we employ a retrieval-augmented model to obtain the context of each sample, thereby strengthening its representation. Comprehensive experiments across benchmark datasets demonstrate that BFPM significantly outperforms state-of-the-art methods, simultaneously improving fairness while maintaining or enhancing performance.

Keywords: Fairness · Pre-processing · Retrieval-augmented Model

1 Introduction

With the widespread application of machine learning in various fields [18, 19], its decisions affect all aspects of society, such as employment [20], education [11], finance [7], and healthcare [23]. However, the decisions made by machine learning models often have potential bias against specific groups. For example, a recent advertising recommendation algorithm demonstrates that different gender groups exhibit distinct performance patterns. Specifically, the model tends to recommend more job advertisements related to the fields of science, technology, engineering, and mathematics to men rather than to women [17]. Another example is the recidivism prediction software used by American courts, which assigns higher risk scores to African Americans while considering whites with similar characteristics to have lower risks [12]. Discrimination entails partitioning data

into distinct groups based on specific attributes, with each group receiving differential treatment. These attributes, such as race and gender, are termed sensitive attributes and are usually predefined.

Data bias is one of the major sources of bias. During manual data collection, biases may originate from multiple sources. First, the preferences and subjective judgments of data collectors may inadvertently introduce bias. For example, data collectors may consciously or unconsciously select certain types of samples while ignoring others. Second, inadequate data representativeness presents another significant concern. When datasets severely underrepresent or completely exclude certain groups while overrepresenting others, models may develop incomplete pattern recognition capabilities. This limitation can cause models to learn and amplify existing biases, potentially exacerbating social inequities [1].

To mitigate data bias, numerous approaches have been proposed from various perspectives. Some of them modify the attribute values of data to alleviate the impact of sensitive attributes on model prediction [2, 6, 27]. In addition, there are also some works that adjust data distribution through sampling techniques [3, 4], or generate different weights for each sample [14]. These methods achieve excellent results, but they have the following shortcomings:

- The dataset may encompass multiple sensitive attributes. However, most existing methods concentrate solely on a single sensitive attribute, thereby neglecting the interactions between multiple sensitive attributes.
- Existing methods for enhancing model fairness often result in a substantial decline in model performance.

To address these issues, in this paper we propose a method called *Balancing Fairness and Performance Architecture Under Multiple Sensitive Attributes* (BFPM). In data pre-processing, we propose a K-nearest neighbor (KNN) [5] based data augmentation method to identify similar neighbors for each sample. Synthetic samples are generated via weighted perturbation and Gaussian noise injection to balance the distribution of sensitive attributes and adjust data ratios for consistency across multiple sensitive attributes. In the in-processing stage, a retrieval-augmented model is introduced to improve sample representation. The model selects similar samples as contextual information using a retrieval mechanism and weighted L2 distance, optimizing prediction accuracy and fairness by combining its own predictions with information from similar samples.

We evaluate fairness with four established metrics and utility with five standard performance indicators. Experiments on three widely-used benchmark datasets show that, relative to state-of-the-art baselines, BFPM achieves superior fairness improvements while maintaining model performance.

In summary, our principal contributions are as follows:

- We propose BFPM that takes into account the interactions between multiple sensitive attributes, thereby addressing the limitations of prior research.
- BFPM works simultaneously in the pre-processing stage and in-processing stage, balancing group fairness and performance.

- We employ a diverse set of metrics to conduct a comprehensive evaluation of the model, and the efficacy of BFPM is empirically validated through experimentation.

2 Related Work

2.1 Fairness Machine Learning

Prior bias removal methods of machine learning can be divided into three categories according to the implementation stage:

Pre-processing algorithms. Pre-processing algorithms aim to mitigate discrimination in datasets by modifying data prior to model training. Feldman et al. [6] quantify disparate impact through classification accuracy metrics and propose a disparate impact remover that modifies attribute distributions to enhance group fairness. Romano et al. [21] substitute original sensitive attributes with label-independent values to train models that approximate equalized odds. Chakraborty et al. [4] identify and remove biased data points that could propagate discrimination in machine learning models. Chakraborty et al. [3] also develop Fair-SMOTE, which oversamples minority group instances through synthetic data generation to achieve balanced class distributions.

In-processing algorithms. These methods are optimized during model training to mitigate biases. Kamishima et al. [16] incorporate a discrimination-aware regularizer into the learning objective to obtain fair classifiers. Zhang et al. [28] employ adversarial learning to prevent inference of sensitive attributes from model predictions, thereby reducing biases. Yang et al. [25] leverage transfer learning to quantify and minimize second-order statistical distances between prediction distributions.

Post-processing algorithms. This method modify model predictions to reduce discrimination. Kamiran et al. [13] propose a decision tree adjustment method that identifies and modifies the most discriminatory leaf labels post-training to produce unbiased classifiers. Hardt et al. [10] develop an equalized odds-based approach that redistributes misclassification costs from disadvantaged groups to decision-makers. Kamiran et al. [15] introduce instance prioritization, focusing on uncertain decision boundary cases by favoring privileged group instances over unprivileged ones.

2.2 Retrieval-Augmented Model

Retrieval-augmented models represent a hybrid artificial intelligence architecture that combines retrieval and generation methods. These models first retrieve contextual information from the knowledge base (typically using vector similarity search or other retrieval techniques) and then integrate this information with the input query through concatenation or attention mechanisms before generating the final output, as shown in Fig. 1. This approach can enhance traditional

machine learning methods by integrating external knowledge, thereby improving predictive performance and generalization ability. Its most well-known application is RAG [8] in large language models (LLMs), which converts user queries into vector representations, retrieves the most relevant documents, combines the retrieved documents with the query, and inputs them into the LLM to enhance the accuracy and richness of generated answers. In addition, retrieval-augmented models can be widely applied in fields such as computer vision [26], and recommendation systems [24].

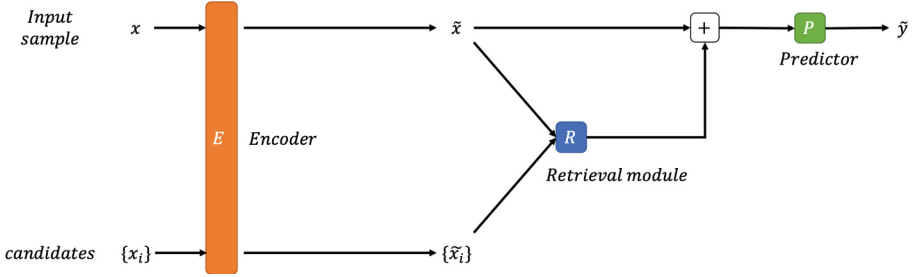


Fig. 1. Retrieval-augmented Model work flow.

Retrieval-augmented models have demonstrated their potential in addressing the challenges associated with tabular data problems. TabR [9] is a method designed to enhance the performance of machine learning models on tabular datasets. TabR is essentially a feedforward network that incorporates a custom KNN component. The model consists of three main components: encoder E, retrieval module R, and predictor P. The encoder E maps the features of the input object to an intermediate representation. The retrieval module R retrieves the most relevant objects based on the similarity module, aggregates the information of these objects, and finally, the predictor P makes predictions based on this aggregated information. Our work builds upon TabR, which is described in detail in Sect. 3.2.

3 Approach

In this section, we provide a detailed description of our approach. First, we elucidate how our data pre-processing method enhances fairness. Next, we employ a retrieval-augmented model to augment various performance metrics of the model. Finally, we integrate data pre-processing and the retrieval-augmented model to illustrate the functionality of our approach.

3.1 Data Pre-processing

Inspired by Fair-SMOTE [3], we propose that imbalanced data distributions may engender model discrimination. To mitigate this, we augment the training data

Algorithm 1. Get Neighbors and Weights**Input:** training dataset *data*, K-Nearest Neighbors model *knn***Output:** *selected_sample*, *neighbors*, *weights*

```

1: random_index  $\leftarrow$  random(0, |data| - 1)
2: selected_sample  $\leftarrow$  data[random_index]
3: distances, indices  $\leftarrow$  knn(selected_sample,
   n_neighbors = 3, return_distance = true)
4: raw_weights  $\leftarrow$   $\frac{1}{\text{distances} + 1e-5}$ 
5: weights  $\leftarrow$   $\frac{\text{raw\_weights}}{\sum \text{raw\_weights}}$ 
6: return selected_sample, data[indices], weights

```

until every joint configuration of sensitive attributes and class labels is equally represented. Our work builds upon Fair-SMOTE, with enhancements made to accommodate multiple sensitive attributes.

Initially, we need to obtain the neighbors of the sample as suitable candidate samples. Algorithm 1, a crucial component of the sample generation algorithm, aims to select a parent candidate from the dataset and find its nearest neighbors using the KNN model. The steps involved are as follows: Initially, a random index is generated to select a sample from the dataset, referred to as the parent candidate. Subsequently, the KNN model is used to find the three nearest neighbors of this parent candidate. The distances between the parent candidate and its neighbors are calculated, and the inverse of these distances is used to compute weights. These weights are then normalized to sum to one, ensuring that they can be used to perform weighted averaging in subsequent steps of the sample generation process.

This function ensures that the selected parent candidate and its nearest neighbors, along with the calculated weights, are returned. These outputs are then utilized to generate new samples, ensuring that the new samples are representative of the local distribution of the data.

Our method involves generating new samples from candidate samples to achieve the effect of expanding the dataset. In Algorithm 1, we detail the process of using KNN to obtain a sample, its three nearest neighbors, and their corresponding weights. Algorithm 2 aims to generate new samples to enhance the diversity and robustness of the dataset. For each feature, specific processing is performed based on its type (boolean, string, or numerical): Boolean features are flipped based on a crossover probability and may be randomly flipped with a certain probability; string features are randomly chosen from the parent candidate and its neighbors based on the calculated weights; numerical features are generated by computing the weighted average and adding Gaussian noise to the difference between the parent candidate and the average.

Finally, we combine the generated new data with the old data to organize them. The goal is to keep the proportion of positive and negative samples in the final dataset the same for each sensitive attribute. Specifically, we first ensure that the original data exists in the new dataset, which helps maintain the model

Algorithm 2. Generate New Sample**Input:** Number of samples n , training dataset $data$ **Output:** Augmented Data $final_data$

```

for each column  $col$  in  $parent\_candidate$  do
  if  $parent\_candidate[col]$  is Boolean then
    if  $cr < \text{random value}$  then
       $value \leftarrow parent\_candidate[col]$ 
    else
       $value \leftarrow \neg parent\_candidate[col]$ 
    end if
    if  $randomvalue < 0.05$  then
       $value \leftarrow \neg value$ 
    end if
    Append  $value$  to  $new\_candidate$ 
  else if  $parent\_candidate[col]$  is String then
     $new\_value \leftarrow \text{Weighted random choice from}$ 
     $\{parent\_candidate[col], neighbors[0][col], neighbors[1][col], neighbors[2][col]\}$ 

    Append  $new\_value$  to  $new\_candidate$ 
  else
     $weighted\_average \leftarrow \text{Weighted average of } neighbors \text{ values for } col$ 
     $new\_value \leftarrow parent\_candidate[col]$ 
     $+ f \times (weighted\_average - parent\_candidate[col])$ 
    Add Gaussian noise to  $new\_value$ 
    Append  $new\_value$  to  $new\_candidate$ 
  end if
end for
Append  $new\_candidate$  to  $total\_data$ 
 $final\_data \leftarrow \text{Convert } total\_data \text{ to DataFrame with original column names}$ 
return  $final\_data$ 

```

performance in subsequent steps. Next, we use a random probability to decide whether the new sample needs to be added to the final dataset and determine whether the dataset is balanced under multiple sensitive attributes. If the dataset at this time is already balanced, then use this dataset as the final dataset, otherwise repeat the previous random process. The search for a new dataset is based on the following formula:

$$X^f =_{balance} X^{(old)} + \sum_{i=0}^m Rand(0, 1) * x_i^{(new)} \quad (1)$$

wherein, X^f represents the final data set, $X^{(old)}$ represents the original data set, m represents the number of new samples generated by the previous algorithm, and $x_i^{(new)}$ represents the i -th new sample. The result of the *balance* method is used as the final data set.

It is important to note that Fair-SMOTE operates on a single sensitive attribute per execution. For example, the German dataset contains two sensitive

attributes: sex and race. When applied, Fair-SMOTE balances the population samples either by sex and label or by race and label, but it does not account for potential interactions between sex and race. In comparison, our proposed method partitions the data into finer subgroups by considering sex, race, and label simultaneously, then equalizes the sample sizes across all resulting groups to ensure fairness.

3.2 Data In-processing

We enhance model fairness by balancing the data distribution in the data pre-processing phase, thereby preserving the flexibility of downstream modeling techniques. In the in-processing stage, we employ a retrieval-based model to optimize model performance without fairness constraints.

Our model is based on TabR proposed by Yury et al. [9] and we make slight modifications to adapt it to the debiased data from our pre-processing stage. Our model comprises three main components: an encoder, a retrieval module, and a predictor.

Encoder. The encoder is responsible for converting the input tabular data into a high-dimensional feature vector. It comprises an input module and several layers of nonlinear transformations. The input module normalizes and one-hot encodes the original features, a step that has already been completed in the preceding debiasing pre-processing method. The nonlinear transformations extract high-order features of the input data through multiple layers of linear transformations and activation functions, while batch normalization is employed to accelerate convergence and enhance model stability.

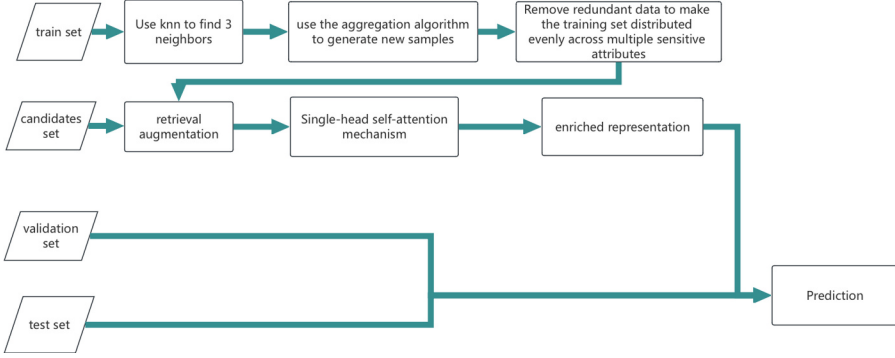
Retrieval Module. The retrieval module serves as the architectural centerpiece of the model. It computes similarity metrics between the target instance and candidate instances in the feature space, subsequently selecting the most similar objects to form a contextual representation. This retrieved context is then integrated with the target’s original representation to produce an enhanced feature vector.

Predictor. The predictor processes the enhanced representation of the target object and produces the final prediction. It comprises multiple layers of linear transformations and nonlinear activation functions.

First, the input sample x is encoded by encoder E to get the representation $\tilde{x} = E(x)$. Similarly, each sample x_i in the candidate dataset I_{cand} is encoded by the same encoder to get $\tilde{x}_i = E(x_i)$. Then the similarity between the target sample and the candidate samples is calculated as follows:

$$S(\tilde{x}, \tilde{x}_i) = -\|W_K(\tilde{x}) - W_K(\tilde{x}_i)\|^2 \quad (2)$$

Here, W_K is a linear transformation matrix, and the formula calculates the L2 distance and adds a negative sign to indicate similarity. Next, the softmax function is used to normalize the similarities and obtain the neighbor weights:

**Fig. 2.** Workflow of BFPM.

$$w_i = \frac{\exp(S(\tilde{x}, \tilde{x}_i))}{\sum_j \exp(S(\tilde{x}, \tilde{x}_j))} \quad (3)$$

Finally, the representation of the target sample is enhanced by aggregating the contributions of the neighbor samples:

$$\tilde{x}' = \tilde{x} + \sum_i w_i V(\tilde{x}_i, y_i) \quad (4)$$

Here, $V(\tilde{x}_i, y_i)$ transforms the information of the candidate samples into an enhanced representation for the target sample, and y_i is the label of the neighbor sample. The enhanced representation $\tilde{x}'$ is then fed into the predictor P to generate the final prediction:

$$\hat{y} = P(\tilde{x}'), \quad P: \mathbb{R}^d \rightarrow Y \quad (5)$$

where Y is the space of the target variable, such as the set of classes in a fair classification task. The complete computational process of the retrieval-augmented model is as follows:

$$\hat{y} = P(E(x) + R(E(x), E(I_{\text{cand}}), Y_{\text{cand}})) \quad (6)$$

where I_{cand} is the candidate dataset, and Y_{cand} is its corresponding labels. The candidate dataset is a balanced dataset generated through pre-processing. R retrieves nearest neighbor samples and adjusts the target sample's representation based on their information and P generates the final prediction.

3.3 Workflow of BFPM

Figure 2 illustrates our overall workflow. During data pre-processing, we first identify three nearest neighbors using the KNN algorithm, then apply distinct aggregation methods to generate synthetic samples. We enforce balance across

multiple sensitive attributes in the final dataset. Specifically, during synthetic data generation, we employ a probabilistic sampling approach, where each generated sample is randomly selected for inclusion in the final dataset. This process iterates until all specified sensitive attributes achieve balanced representation.

Next, in the in-processing stage, we utilize the L2 distance to compare and obtain sample context and employ a retrieval-augmented model to achieve a richer feature representation. Training is halted via early stopping as soon as validation performance begins to degrade while training loss continues to decline, thereby mitigating overfitting. Model accuracy and fairness are ultimately assessed on the test set.

4 Experiment Setup

In this section, we present our experimental setup, detailing the datasets employed, fairness metrics, performance metrics, and baseline used for comparison.

4.1 Datasets

We select the Adult, Compas, and German Credit datasets for our experiments, as they are widely adopted benchmarks in fair machine learning research and each contains multiple sensitive attributes. Below we provide detailed descriptions of these datasets:

Adult. The Adult dataset, also known as the Census Income dataset. It is used to predict whether an individual’s annual income exceeds \$50,000 based on 14 attributes. The dataset contains 48,842 samples. We selected **sex**, **race**, and **age** as sensitive attributes. Sex and race are binary, and for ease of calculation, we binarize the age attribute using a threshold of 35 years, labeling individuals aged 35 and above as 1, and those below 35 as 0.

COMPAS. The COMPAS dataset comprises criminal history records of defendants and is widely used for recidivism risk prediction. The dataset contains 7,214 samples, each characterized by 28 attributes. Similar to the Adult dataset, we designate **sex**, **race**, and **age** as sensitive attributes. We binarize the age attribute using a threshold of 35 years.

German Credit. The German Credit dataset it is used for credit assessment, aiming to predict whether an individual poses a credit risk. The dataset includes a total of 1,000 samples, each with 20 attributes. For this dataset, we selected **sex** and **age** as sensitive attributes to supplement the previous datasets and further validate the effectiveness of our method. The age attribute is also binarized.

4.2 Metrics

We introduce the five performance metrics and four fairness metrics used in our experiments. Each metric is defined with a brief explanation and its corresponding formula.

Performance Metrics. We focus on binary classification tasks, where sample labels are categorized into positive and negative classes. True Positive (TP) refers to the number of correctly predicted positive sample, whereas True Negative (TN) refers to the number of correctly predicted negative sample. False Positive (FP) indicates the number of negative samples that are incorrectly predicted as positive, and False Negative (FN) indicates the number of positive samples that are incorrectly predicted as negative.

Recall. measures the proportion of actual positives that are correctly identified by the model. It is defined as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (7)$$

False Alarm. measures the proportion of actual negatives that are incorrectly identified as positives. It is defined as:

$$\text{False Alarm} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (8)$$

Precision. measures the proportion of predicted positives that are correctly identified. It is defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (9)$$

Accuracy. measures the proportion of all correct predictions (both positives and negatives) out of the total predictions. It is defined as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (10)$$

F1 Score. is the harmonic mean of precision and recall, providing a balance between the two. It is defined as:

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (11)$$

Fairness Metrics. Our research focuses on *group fairness*, which aims to ensure equal treatment of privileged and non-privileged groups based on sensitive attributes. Below are the metrics utilized in this paper. Let $\hat{Y}$ is the prediction of model, Y is the label of sample. A is the sensitive attribute and $A \in \{0, 1\}$.

Average Odds Difference (AOD). measures the difference in the true positive rates and false positive rates between different groups. It is defined as:

$$\begin{aligned} \text{AOD} = \frac{1}{2} \Big(& |P(\hat{Y} = 1 \mid Y = 1, A = 0) - P(\hat{Y} = 1 \mid Y = 1, A = 1)| \\ & + |P(\hat{Y} = 1 \mid Y = 0, A = 0) - P(\hat{Y} = 1 \mid Y = 0, A = 1)| \Big) \end{aligned} \quad (12)$$

Equal Opportunity Difference (EOD). measures the difference in true positive rates between different groups. It is defined as:

$$\text{EOD} = |P(\hat{Y} = 1 \mid Y = 1, A = 0) - P(\hat{Y} = 1 \mid Y = 1, A = 1)| \quad (13)$$

Statistical Parity Difference (SPD). measures the difference in positive outcome rates between different groups. It is defined as:

$$\text{SPD} = |P(\hat{Y} = 1 \mid A = 0) - P(\hat{Y} = 1 \mid A = 1)| \quad (14)$$

Disparate Impact (DI). measures the ratio of the positive outcome rates between different groups. It is defined as:

$$\text{DI} = \left| 1 - \frac{P(\hat{Y} = 1 \mid A = 0)}{P(\hat{Y} = 1 \mid A = 1)} \right| \quad (15)$$

In order to study the fairness performance of multiple sensitive attributes, we introduce the intersectional fairness metrics. Now $A = A_1 \times A_2 \times \dots \times A_n$, $A_i \in \{0, 1\}$ is the i -th sensitive attribute. S represents the cross combination set of all sensitive attributes, and $A = s$ represents the specific sensitive attribute combination. We take SPD as an example, and other fairness metrics are defined in a similar way:

$$\text{Inter_SPD} = \max_{s \in S} P(\hat{Y} = 1 \mid A=s) - \min_{s \in S} P(\hat{Y} = 1 \mid A=s) \quad (16)$$

These metrics jointly furnish a comprehensive evaluation of the model, enabling us to quantify the performance–fairness trade-off in its predictions.

4.3 Baseline

We employ three baseline models: a Multilayer Perceptron (MLP) serves as a performance benchmark. Since our debiasing occurs during the pre-processing stage, we select two pre-processing methods—Fairway and Fair-SMOTE—to facilitate comparisons of BFPM’s improvements in fairness.

MLP. [22] is a type of feedforward artificial neural network that consists of multiple layers of neurons. As a benchmark classifier that does not incorporate specific fairness adjustments, MLP may exhibit certain biases in scenarios involving sensitive attributes.

Fairway. [4] is an under-sampling method that identifies and removes ambiguous (biased) data points in the training data, which may lead to bias. This is achieved by training two models based on privileged and non-privileged groups, respectively. Additionally, Fairway adjusts model parameters through multi-objective optimization to further reduce bias.

Fair-SMOTE. [3] is a bias-corrected variant of SMOTE that synthesizes minority-class instances to equalize class distributions and enhance predictive fairness.

Table 1. Fairness comparison of BFPM with prior algorithms in individual sensitive attribute on three datasets. Bold indicates the first rank, and underlined in the metrics indicates the second rank.

Dataset	Sensitive Attribute	Algorithm	AOD (-)	EOD (-)	SPD (-)	DI (-)
Adult	Sex	MLP	0.09	0.25	0.17	0.87
		Fairway	0.02	0.03	<u>0.11</u>	0.38
		Fair-SMOTE	0.02	0.06	<u>0.13</u>	<u>0.34</u>
		BFPM	<u>0.03</u>	<u>0.05</u>	0.09	0.26
	Race	MLP	0.06	0.16	0.09	0.6
		Fairway	<u>0.02</u>	<u>0.03</u>	<u>0.06</u>	0.32
		Fair-SMOTE	0.03	<u>0.03</u>	0.08	<u>0.22</u>
		BFPM	0.01	0.01	0.05	0.16
	Age	MLP	0.15	0.4	0.24	0.93
		Fairway	<u>0.04</u>	0.11	<u>0.15</u>	0.35
		Fair-SMOTE	0.05	<u>0.07</u>	0.16	<u>0.33</u>
		BFPM	0.01	0.03	0.14	0.27
COMPAS	Sex	MLP	0.07	0.23	0.31	0.37
		Fairway	<u>0.03</u>	0.03	<u>0.06</u>	<u>0.12</u>
		Fair-SMOTE	0.06	<u>0.05</u>	0.04	0.08
		BFPM	0	0.06	0.09	0.2
	Race	MLP	0.05	0.08	0.15	0.22
		Fairway	<u>0.02</u>	<u>0.04</u>	0.07	<u>0.07</u>
		Fair-SMOTE	0.03	0.07	<u>0.06</u>	0.11
		BFPM	0.01	0.03	0.01	0.02
	Age	MLP	0.09	0.45	0.55	0.61
		Fairway	0.04	0.06	<u>0.11</u>	<u>0.15</u>
		Fair-SMOTE	<u>0.02</u>	<u>0.03</u>	<u>0.11</u>	0.21
		BFPM	0	0.01	0.05	0.12
German Credit	Sex	MLP	0.11	0.08	0.06	0.45
		Fairway	<u>0.05</u>	<u>0.04</u>	<u>0.07</u>	<u>0.11</u>
		Fair-SMOTE	<u>0.05</u>	0.05	0.05	0.13
		BFPM	0.02	0.02	0.08	0.07
	Age	MLP	0.16	0.05	0.23	0.42
		Fairway	<u>0.03</u>	0.02	0.03	<u>0.13</u>
		Fair-SMOTE	0.01	<u>0.01</u>	<u>0.04</u>	<u>0.13</u>
		BFPM	0.01	0	0.03	0.11

5 Results

In this section, we present our experimental results and answer three RQs.

5.1 RQ1: How Does BFPM Perform Relative to Baseline Methods in Terms of Fairness?

As illustrated in Table 1, BFPM exhibits significant superiority in fairness metrics across diverse datasets and sensitive attributes when compared to alternative methods. On the Adult dataset, BFPM shows outstanding performance. For the Race attribute, it ranks first in all metrics (AOD, EOD, SPD, DI), outperforming MLP, Fairway, and Fair-SMOTE. For the Sex attribute, it ranks second in all metrics. For the Age attribute, it ranks second in AOD and SPD, first in EOD, and second in DI.

The COMPAS dataset further underscores BFPM’s performance. For the Race attribute, it ranks first in SPD and DI, and second in AOD and EOD. For the Sex attribute, it ranks second in all metrics. For the Age attribute, it ranks second in AOD and SPD, first in EOD, and second in DI.

Similarly, on the German Credit dataset, BFPM maintains its strong performance. For the Sex attribute, it ranks second in all metrics. For the Age attribute, it ranks second in AOD and SPD, first in EOD, and second in DI.

In summary, BFPM achieves the lowest AOD values in 7 out of 8 comparisons, the lowest EOD values in 6 out of 8 comparisons, and the lowest SPD values in 6 out of 8 comparisons. Additionally, BFPM performs the best in 7 out of 8 DI calculations, significantly outperforming other methods. Its frequent first or second place rankings in fairness metrics demonstrate its superior effectiveness in ensuring fairness compared to other methods.

Table 2. Performance comparison of BFPM with prior algorithms on three datasets. Bold indicates the first rank, and underlined in the metrics indicates the second rank.

Dataset	Algorithm	Recall (+)	False Alarm (-)	Precision (+)	Accuracy (+)	F1 score (+)
Adult	MLP	0.40	<u>0.05</u>	0.71	<u>0.82</u>	0.51
	Fairway	0.25	0.04	<u>0.70</u>	0.72	0.37
	Fair-SMOTE	<u>0.73</u>	0.15	0.49	0.76	<u>0.59</u>
	BFPM	0.75	0.22	0.67	0.83	0.71
COMPAS	MLP	0.69	0.44	0.67	<u>0.63</u>	0.68
	Fairway	0.55	<u>0.21</u>	0.58	0.58	<u>0.56</u>
	Fair-SMOTE	0.57	0.32	<u>0.68</u>	0.62	0.62
	BFPM	<u>0.62</u>	0.07	0.75	0.66	0.68
German Credit	MLP	0.94	0.81	<u>0.72</u>	0.76	0.82
	Fairway	<u>0.78</u>	0.76	0.68	0.65	0.73
	Fair-SMOTE	0.61	0.38	0.65	0.66	0.63
	BFPM	0.74	<u>0.45</u>	0.77	<u>0.70</u>	<u>0.75</u>

5.2 RQ2: Does BFPM Improve Performance Relative to the Baseline Methods?

As shown in Table 2, BFPM generally achieves higher or comparable performance across all datasets. On the Adult dataset, BFPM leads in recall, accuracy, and F1 score. On the COMPAS dataset, BFPM has the lowest false alarm rate, the highest precision, accuracy and F1 score, but ranks second in Recall. On the German Credit dataset, BFPM achieves the highest precision, but ranks second in false alarm rate, accuracy and F1 score.

Based on the experimental results from three datasets, BFPM delivers optimal or suboptimal performance across multiple key metrics. Specifically, in 15 comparisons (3 datasets $\times$ 5 metrics), BFPM achieves 8 best performances and 4s-best performances, significantly outperforming Fairway and Fair-SMOTE. Note that BFPM often surpasses MLP in certain metrics, proving its ability to enhance the original model’s performance in specific scenarios.

5.3 RQ3: How Does BFPM Perform with Multiple Sensitive Attributes?

Table 3. Fairness comparison of BFPM with prior algorithms in multiple sensitive attributes on Adult dataset. Bold indicates the first rank, and underlined in the metrics indicates the second rank.

Metrics	Algorithm	Sex-race-age	Sex-race	Race-age	Sex-age
Inter_AOD	MLP	0.15	0.40	0.24	0.39
	Fairway	<u>0.04</u>	<u>0.11</u>	0.18	0.32
	Fair-SMOTE	0.05	0.17	<u>0.16</u>	<u>0.31</u>
	BFPM	0.01	0.03	0.14	0.16
Inter_EOD	MLP	0.09	0.25	0.17	0.87
	Fairway	0.02	0.03	<u>0.11</u>	0.38
	Fair-SMOTE	0.02	0.06	0.13	<u>0.34</u>
	BFPM	<u>0.03</u>	<u>0.05</u>	0.09	0.26
Inter_SPD	MLP	0.06	0.16	0.09	0.39
	Fairway	<u>0.02</u>	<u>0.03</u>	<u>0.06</u>	0.32
	Fair-SMOTE	0.03	<u>0.03</u>	0.05	<u>0.31</u>
	BFPM	0.01	0.01	0.05	0.16
Inter_DI	MLP	0.15	0.40	0.24	0.93
	Fairway	<u>0.04</u>	<u>0.11</u>	0.18	0.37
	Fair-SMOTE	0.05	0.17	<u>0.16</u>	<u>0.33</u>
	BFPM	0.01	0.03	0.14	0.27

Taking the Adult dataset as an example, the results in Table 3 show that BFPM achieved the best performance in 14 fairness comparisons, significantly outperforming MLP, Fairway, and Fair-SMOTE methods. BFPM performs the best in all sensitive attribute combinations on the three indicators of Inter_AOD, Inter_SPD, and Inter_DI, indicating that this method can effectively reduce the unfairness of the model on multiple sensitive attribute. In addition, BFPM also achieved twice the best among the four sensitive attribute combinations in the Inter_EOD metric,

One reason is that our method ensures that the proportions of multiple sensitive attributes in the dataset remain consistent, while the Fair-SMOTE method only pre-processes a sensitive attribute. Another reason is that our method can generate more data for debiasing, and we have improved the way of generating data to make the new data closer to the real data.

6 Conclusion and Future Work

In this paper, we propose a method called BFPM that integrates the pre-processing and in-processing stages. During the pre-processing stage, we develop an enhanced KNN algorithm that generates synthetic samples to balance fine-grained subpopulations according to both sensitive attributes and labels, thereby ensuring equitable data distribution. For the in-processing phase, we introduce a retrieval-augmented model that optimizes sample representations to simultaneously enhance model utility while maintaining fairness. Comprehensive experimental results show that BFPM attains a superior performance–fairness trade-off under multiple sensitive attributes.

A potential future direction is to merge these two stages and employ a unified method to enhance both fairness and performance concurrently. This approach could streamline the process and decrease computational time. Furthermore, our current method is validated on tabular datasets. Extending its applicability to other data modalities, such as images and text, would represent a valuable avenue for future investigation.

References

1. Birhane, A.: Algorithmic injustice: a relational ethics approach. *Patterns* **2**(2) (2021)
2. Calmon, F.P., Wei, D., Ramamurthy, K.N., Varshney, K.R.: Optimized data pre-processing for discrimination prevention (2017)
3. Chakraborty, J., Majumder, S., Menzies, T.: Bias in machine learning software: Why? how? what to do? In: *Proceedings of the 29th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, pp. 429–440 (2021). <https://doi.org/10.1145/3468264.3468537>, <http://arxiv.org/abs/2105.12195>
4. Chakraborty, J., Majumder, S., Yu, Z., Menzies, T.: Fairway: a way to build fair ml software. In: *Proceedings of the 28th ACM joint meeting on European software engineering conference and symposium on the foundations of software engineering*, pp. 654–665 (2020)
5. Cover, T., Hart, P.: Nearest neighbor pattern classification. *IEEE Trans. Inf. Theory* **13**(1), 21–27 (1967). <https://doi.org/10.1109/TIT.1967.1053964>
6. Feldman, M., Friedler, S., Moeller, J., Scheidegger, C., Venkatasubramanian, S.: Certifying and removing disparate impact. *ACM* (2014)
7. Furman, J., Seamans, R.: Ai and the economy. *Innov. Policy Econ.* **19**(1), 161–191 (2019)
8. Gao, Y., et al.: Retrieval-augmented generation for large language models: a survey. *arXiv preprint arXiv preprint arXiv:2312.10997*, **2** 1 (2023)
9. Gorishniy, Y., et al.: TABR: tabular deep learning meets nearest neighbors in 2023 (2023). <http://arxiv.org/abs/2307.14338>
10. Hardt, M., Price, E., Srebro, N.: Equality of opportunity in supervised learning. In: *Proceedings of the 30th International Conference on Neural Information Processing Systems (NeurIPS 2016)*, pp. 3315–3323 (2016)
11. Holmes, W., Tuomi, I.: State of the art and practice in ai in education. *Eur. J. Educ.* **57**(4), 542–570 (2022)

12. Julia Angwin, Jeff Larson, S.M., Kirchner, L.: Machine bias: there's software used across the country to predict future criminals. and it's biased against blacks. (2016). <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
13. Kamiran, F., Calders, T.: Classifying without discriminating. In: 2009 2nd international conference on computer, control and communication, pp. 1–6. IEEE (2009)
14. Kamiran, F., Calders, T.: Data preprocessing techniques for classification without discrimination. *Knowl. Inf. Syst.* **33**(1), 1–33 (2012)
15. Kamiran, F., Karim, A., Zhang, X.: Decision theory for discrimination-aware classification. In: 2012 IEEE 12th international conference on data mining, pp. 924–929. IEEE (2012)
16. Kamishima, T., Akaho, S., Asoh, H., Sakuma, J.: Fairness-aware classifier with prejudice remover regularizer. In: Machine Learning and Knowledge Discovery in Databases: European Conference, ECML PKDD 2012, Bristol, UK, September 24–28, 2012. Proceedings, Part II 23, pp. 35–50. Springer (2012)
17. Lambrecht, A., Tucker, C.: Algorithmic bias? an empirical study of apparent gender-based discrimination in the display of stem career ads. *Management Science* (2019)
18. Li, Y., et al.: SRDK: towards efficient 3d point cloud segmentation via structure- and relation-aware knowledge distillation. arXiv preprint [arXiv:2506.17290](https://arxiv.org/abs/2506.17290) (2025)
19. Li, Y., et al.: Frequency-aligned knowledge distillation for lightweight spatiotemporal forecasting. [arXiv:2507.02939](https://arxiv.org/abs/2507.02939) (2025)
20. Petropoulos, G.: The impact of artificial intelligence on employment. *Praise for Work in the Digital Age* **119**, 121 (2018)
21. Romano, Y., Bates, S., Candes, E.: Achieving equalized odds by resampling sensitive attributes. *Adv. Neural. Inf. Process. Syst.* **33**, 361–371 (2020)
22. Rumelhart, D.E., Hinton, G.E., Williams, R.J.: Learn. representations by back-propagating errors. *nature* **323**(6088), 533–536 (1986)
23. Shaheen, M.Y.: Applications of artificial intelligence (ai) in healthcare: a review. *ScienceOpen Preprints* (2021)
24. Xu, J., et al.: Rallrec: Improving retrieval augmented large language model recommendation with representation learning. arXiv preprint [arXiv:2502.06101](https://arxiv.org/abs/2502.06101) (2025)
25. Yang, J., Zhang, L., Zhang, M.: Making fair classification via correlation alignment. In: ECAI 2024, pp. 842–849. IOS Press (2024)
26. Yasunaga, M., et al.: Retrieval-augmented multimodal language modeling. arXiv preprint [arXiv:2211.12561](https://arxiv.org/abs/2211.12561) (2022)
27. Zemel, R., Wu, Y., Swersky, K., Pitassi, T., Dwork, C.: Learning fair representations. *JMLR.org* (2013)
28. Zhang, B.H., Lemoine, B., Mitchell, M.: Mitigating unwanted biases with adversarial learning. In: Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society, pp. 335–340 (2018)